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RESEARCH-ARTICLE

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LaMHA: Efficient Multimodal Malicious Meme Classification via LoRA-Tuned Adaptation and Attentive-MLP Fusion

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Abstract

To address insufficient cross-modal semantic alignment and limited dynamic discrimination in multi-modal malicious meme classification, this paper proposes a lightweight architecture integrating Low-Rank Adaptation (LoRA) and Attention-guided Multilayer Perceptron (aMLP). The architecture achieves efficient and accurate malicious meme classification through a two-stage cascade: 1) The LoRA module embedded in CLIP's text encoder performs lightweight text feature projection via low-rank matrix decomposition, enabling efficient cross-modal adaptation with only 0.3% fine-tuned parameters; 2) The aMLP module at the classification layer dynamically fuses multi-modal features using cross-modal attention weights and suppresses noise via a gating mechanism. Experiments on HMC and HarMeme datasets demonstrate superior performance in Accuracy (77.74% / 83.72%), F1 (69.79% / 79.82%), and AUROC (84.01% / 92.16%), with a notable 4.33% AUROC gain on cultural metaphors. Ablation studies confirm nonlinear synergistic effects (11.53% F1 improvement on HMC). This work provides an efficiency-precision balanced solution for multi-modal content safety.

CCS Concepts

• **Computing methodologies** → Artificial intelligence; Natural language processing; Artificial intelligence; Computer vision.

Keywords

Deep Learning, Multi-modal Fusion, Meme Classification, Multi-modal Analysis

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1 INTRODUCTION

With the rapid development of the Internet and the continuous increase of network bandwidth, multimodal data has become the main carrier of information dissemination on social media due to

its rich forms of expression [1]. Among them, as a unique form of multimodal expression, memes [2] are different from traditional text messages or single emojis, can convey rich information with concise images [3]. They are usually images that contain images along with text pieces that are artificially superimposed in or around the images, expressing a range of concepts and emotions, such as humor, kindness and irony, etc. [4].

Since meme images typically take the form of “image with text”, they can be quickly imitated and spread if they can evoke associations in people’s visual, linguistic, musical, or behavioral aspects. However, some memes have been created to attack individuals and communities directly or indirectly, based on attributes such as race [5], gender [6], sexual orientation [7], religion, nationality, political affiliation [8], immigration status, and personal beliefs. Not only does this effect people’s online experience, but it can lead to radicalization online and even incite real-world hate crimes [9].

Unfortunately, considering the multimodal nature of memes and the massive growth of information on the Internet, it is almost impossible to conduct manual online moderation [10] and rely on single text modality for keyword filtering. The identification and classification of malicious memes is particularly challenging, as Figure 1 shows, when innocuous images (black monkeys) collocated with different text (“when someone starts telling you a story, and you already know they’re full of shit”/“Obama voters”), the former meme expresses neutral, harmless feelings, while the latter one implies racist, and attacks based on political positions. It is only when the text and the image are fused together that their true intentions are subtly revealed. Therefore, Facing the massive and still real-time updated internet memes, the development of automated malicious meme classification method is extremely necessary.

To tackle the aforementioned task, we propose a malicious meme classification model named LaMHA. In this paper, Related works in this field will be described in section 2. The proposed LaMHA model will be detailed in section 3 while section 4 will present the experimental results and comparative analysis to demonstrate the superior performance of our model; Finally, we summarize our work in section 5.

2 RELATED WORKS

2.1 Meme Classification

Since the meme sentiment analysis task was first proposed at SemEval-2020 [11], more and more meme detection and classification work has begun in 2020 [12] [13].

Facebook spearheaded the “Hateful Memes Challenge”, which aims to detect and neutralize hate speech in multimodal meme content [3]. In the same year, Suryawanshi et al. developed the



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Figure 1: Meme samples taken from the HMC dataset [3].

MultiOFF dataset to detect offensive content in memes [14]. In addition, Keswani et al. mined the emotional information contained in the meme and the category to which it belonged based only on the textual information embedded in the meme [15], while Ling et al. focused on distinguishing the visual elements of viral image memes from non-viral image memes to predict the viral rate of memes [16]. Tanaka et al. also created a humor detection dataset in 2022 and used a machine learning model for detection [17]. Qu introduced a disinformation meme dataset to detect the disinformation that may be hidden and spread in some real-world memes [18]. Bhandari et al. focused on the detection of memes related to the Russia-Ukraine war by creating the Crisis Hate MM dataset [19], and the recent work of Shah et al. [4] expands the scope of multimodal memetic analysis to address more complex expressions, including hatred, hateful targets and humor.

2.2 Multimodal Feature Fusion

With the deepening of research, researchers have found that the meme classification work that relies on unimodal text-embedded images has great limitations [20], and needs to understand the context, and there are ambiguities and subjectivity in it. Subsequent approaches attempt to address this challenge by integrating more complex architectures and fusion mechanisms such as stacking models, early fusion techniques, and glove embedding [21].

Bhat et al. combined object labels and text analysis combined with headings to train a classification model. This approach enables a correct assessment of the visual and textual context that makes up the meme [22]. Fang et al. used an Image Subtitle Supervision (ICS) auxiliary method to counter the problem that conventional algorithms have less understanding of the semantics of image modalities, which uses the image title to better understand the image modality and supervise the feature learning of the image [23].

The early fusion applications of multimodal data are often used in the field of retrieval, such as the use of Latent Dirichlet Allocation (LDA) and SIFT algorithms to represent the features of text and images, respectively, to explore the common modeling problem of multimodal text and images [24], and to learn the feature correlation of two modalities with a higher level of abstraction with the help of Canonical Correlation Analysis (CCA). The use of machine learning

methods [25] [26] and deep learning [27] to fine-tune the model and demonstrate the close connection between text and image has also been shown to be effective.

Considering that the eigenvectors are simply spliced together, the complex relationship between modalities cannot be well modeled [28]. You et al. first fine-tuned the pre-trained CNN model with an image sentiment dataset, then extracted image features, and trained the word2vec model using unsupervised learning, respectively, and experimented with feature-level fusion and decision-level fusion methods, respectively, and achieved better results than single modality [29]

Recent approaches attempt to solve this problem with complex architecture: Hate-CLIPper [30] fine-tuned contrastive language image pre-training (CLIP) [31] projections, while Mei et al. [32] improved the Hate-CLIPper method by aligning embeddings from the same class, dynamically retrieving them, and training them with contrast loss and cross-entropy loss. Burbi et al. [33] also utilizes a clipping model, but unravel the representation before performing text inversion, i.e., using a combinator module to fuse image and text features and pass them to MLP for binary classification.

However, most of these studies rely on simply concatenating feature vectors of different modalities to directly extract global image features, which lacks the association with text information. Secondly, in the process of extracting embedded text information using OCR technology, a certain amount of noise is often introduced, which affects the performance of the model. In addition, the data labeling of related datasets is often subjective, and the distribution of positive and negative samples is unbalanced, which may lead to overfitting of the classification effect in some classes.

To tackle the task, we propose a malicious meme classification model named LaMHA. Experiments show that our model achieves SotA results on the challenging datasets. We summarize our contributions as follows:

- Breaking the Computational Redundancy Challenge in Multimodal Adaptation: A lightweight cross-modal alignment method based on LoRA [34] is proposed to achieve efficient text-visual semantic collaboration without full-parameter fine-tuning, significantly reducing the model deployment threshold.

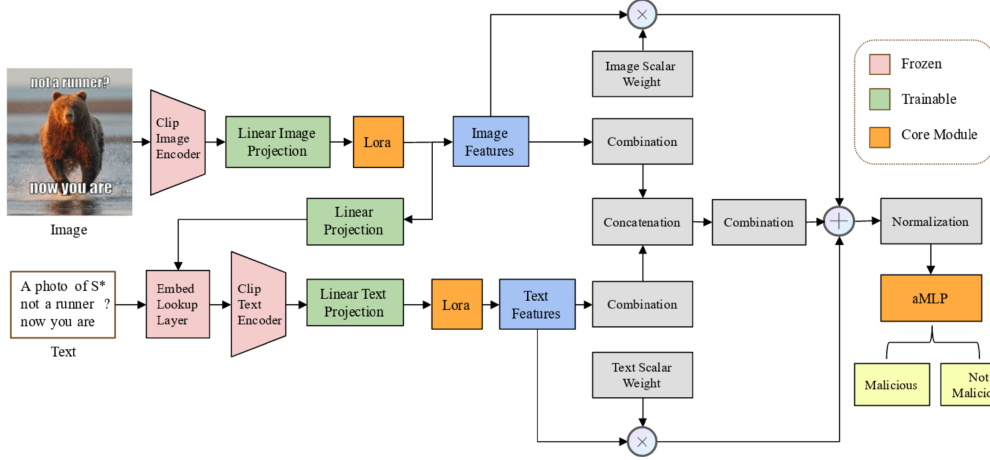


Figure 2: The architecture of the proposed LaMHA model.

- Overcoming the Interaction Limitations of Static Feature Fusion: An attention-guided aMLP [35] module is designed to enhance the parsing ability of cross-modal implicit semantics (such as cultural metaphors and adversarial perturbations) through dynamic weight allocation.
- Establishing a Paradigm of Synergistic Optimization between Efficiency and Accuracy: An end-to-end multimodal detection framework is constructed to achieve real-time inference while maintaining high classification accuracy, providing a feasible solution for edge computing scenarios.

3 PROPOSED METHODS

In this section, we describe our proposed model, LaMHA. This study proposes revolutionary architecture with two core innovations:

Low-Rank Adaptation (LoRA): Integrated after the linear projection layer of CLIP’s text encoder, it achieves parameter-efficient cross-modal alignment via low-rank matrix decomposition. This enables precise text feature projection in low-rank subspaces while preserving CLIP’s original knowledge.

Attention-guided MLP (aMLP): Deployed at the classification layer, it enhances critical semantic correlations through three-stage dynamic fusion (cross-modal attention generation, residual feature recombination, and gated decision-making), while suppressing irrelevant noise. These modules collaboratively form a “feature adaptation to dynamic discrimination” cascade mechanism, overcoming limitations of static fusion without significantly increasing model complexity.

As illustrated in Figure 2, this architecture establishes a new paradigm of lightweight module co-optimization: The LoRA module significantly reduces text encoder tuning parameters through low-rank constrained updates, resolving computational redundancy

and overfitting risks in full-parameter fine-tuning. The aMLP module implements context-aware feature recombination via attention gating. This design maintains CLIP’s knowledge integrity while achieving efficient task adaptation and real-time inference. It provides a scalable technical pathway for multimodal security detection enhancing robustness against cultural metaphors and adversarial samples without stacking complex modules, thereby validating the core value of lightweight dynamic interaction in semantic decoupling.

3.1 Low-Rank Adaptation Fine-tuning (LoRA)

To efficiently adapt the pretrained CLIP text encoder for multimodal malicious meme detection while preserving its semantic knowledge, we implement Low-Rank Adaptation (LoRA) technology. Traditional fine-tuning methods require full-parameter updates of CLIP, which leads to excessive computational resource consumption and overfitting risks. The core innovation of LoRA lies in decomposing the weight update matrix ΔW into a product of low-rank matrices during task adaptation. Specifically, for the linear projection layer weight matrix $W \in \mathbb{R}^{d \times k}$ in CLIP’s text encoder, its update is constrained as $\Delta W = BA^T$, where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ are trainable low-rank matrices with rank $r \ll \min(d, k)$. The modified forward propagation becomes:

$$h = Wx + BA^T x$$

The original weights W remain frozen, with only B and A being optimized. This design reduces trainable parameters from $d \times k$ to $r(d + k)$, significantly lowering computational overhead. In our architecture, LoRA is strategically positioned after the linear projection layer of CLIP’s text encoder. This placement enables direct refinement of text features x within a low-rank subspace, allowing precise alignment with visual semantics. For instance, when processing culturally specific metaphors, LoRA captures such

implicit associations through low-rank mapping while preserving CLIP’s original semantic space integrity.

3.2 Attention-guided Multilayer Perceptron (aMLP)

To enhance dynamic fusion of multimodal features, we introduce an attention-guided multilayer perceptron (aMLP) in the classification layer. Conventional MLPs employ static weights for feature fusion, lacking adaptability to varying semantic focuses across samples. The aMLP framework implements three key operations:

3.2.1 Cross-Modal Attention Generation. Given LoRA-adapted text features $h_t \in \mathbb{R}^d$ and image features $h_v \in \mathbb{R}^d$, we compute attention weights:

$$\begin{aligned} Q &= W_q h_t \\ K &= W_k h_v \\ A &= \text{softmax} \left(\frac{QK^T}{\sqrt{d}} \right) \end{aligned}$$

where $W_q, W_k \in \mathbb{R}^{d \times d}$ are learnable parameters, and $A \in \mathbb{R}^{1 \times n}$ denotes the importance distribution of text-to-image regions (n = number of image patches).

This section aims to generate an attention weight matrix of text and image features via learnable parameters. This matrix acts as a “semantic compass,” quantifying the relevance between text descriptions and image regions. For instance, when the text contains metaphors, W_q and W_k map the subject in the text to the corresponding subject area in the image, A produces high weights (>0.9) in the subject area and the weight in the irrelevant area tends to be 0.

3.2.2 Residual Feature Recombination. The image features are adaptively weighted while preserving original information through residual connections:

$$h'_v = A \cdot h_v + h_v$$

This section is designed to retain the original features when weighing the image features to form an “adaptive filter” mechanism. For instance, for fuzzy violent scenes, the operation highlights key areas (such as armed hands) and preserves the contextual background information (such as ambient light, object position) to prevent over-filtering.

3.2.3 Gated Fusion and Classification. Dynamic blending of text and recombined image features is achieved via:

$$\begin{aligned} g &= \sigma \left(W_g \begin{bmatrix} h_t \\ h'_v \end{bmatrix} \right) \\ h_{\text{out}} &= g \odot (W_f h'_v) + (1 - g) \odot h_t \end{aligned}$$

where σ denotes the sigmoid function, and $W_g \in \mathbb{R}^{2d \times d}$, $W_f \in \mathbb{R}^{d \times d}$ are classification parameters. The final output h_{out} is fed into a softmax layer for hate probability prediction.

This section employs dynamic gating (within the range of 0-1) to control the contribution ratio of text and image features, forming a “semantic switch.” When the image features are explicit, the gating leans towards the visual channel; when the text contains puns (for example, “cleaning up trash” implying ethnic cleansing), it enhances the decision-making weight of the text. The ultimately integrated

features are then output the predicted probability via the Softmax layer.

At the classification stage, aMLP automatically amplifies critical cross-modal correlations. For example, when violent imagery coexists with linguistically ambiguous attacks, aMLP enhances weight for semantically congruent regions while suppressing irrelevant background noise. This mechanism enables effective discrimination between malicious multimodal synergy and harmless coincidences.

4 EXPERIMENTS

4.1 Datasets

In this paper, we primarily evaluated our proposed algorithm on the two most mainstream datasets in the field: the HMC dataset [3] and the HarMeme dataset [36]. Figure 3 shows several examples of these two datasets.

HMC dataset was proposed by Facebook for the Hateful Memes Challenge. The hatred in HMC dataset is aimed mainly at religion, race, disability, and sex. While HarMeme dataset consists of real-world meme images related to topics such as COVID-19 and US politics. It is classified into three groups: highly harmful, partially harmful, and harmless. The dataset is partitioned into training, validation, and test sets in the ratio of 3013:177: 354. Detailed information regarding these datasets can obtain from Table 1.

Malicious Meme, in the datasets used in this paper, is strictly defined as follows: A direct or indirect attack on people based on characteristics, including ethnicity, race, nationality, immigration status, religion, caste, sex, gender identity, sexual orientation, and disability or disease. We define attack as violent or dehumanizing (comparing people to non-human things, e.g., animals) speech, statements of inferiority, and calls for exclusion or segregation. Mocking hate crime is also considered hate speech.

Both datasets ensure the standardization and consistency of the annotation work through the multi-stage annotation work of professional annotators and introduce “benign confounders” to increase the difficulty of the task, ensuring that the model cannot only rely on a single modality (image or text) for classification, and must be combined with multimodal information to succeed. This design not only makes the task more challenging but also brings it closer to the complex real-world situation. Both are widely used in the field of malicious meme classification.

4.2 Validation Metrics

In the task of malicious meme detection, we regard the problem as a binary classification task and employ accuracy (Accuracy), AUROC (Area Under ROC), and F1 value (F1-Score) as the key indicators to evaluate model performance. Specifically, accuracy measures the proportion of overall correct predictions made by the model, calculated by the formula:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Where TP (True Positive) denotes the number of correctly identified malicious memes, TN (True Negative) represents correctly classified non-malicious samples, FP (False Positive) counts benign content misclassified as malicious, and FN (False Negative) indicates



Figure 3: Meme samples from the HMC dataset and the HarMeme dataset.

Table 1: Summary of datasets used in the experiment.

Dataset	Size	Multimodal	Sub-Classes	Annotation	Source	Theme
HarMeme	3,544	✓	✓	2 period	Google	COVID-19, U.S. Politics
HMC	10,000	✓	✓	4 period	Facebook, Twitter, Instagram	General Discourse

undetected malicious instances. While Accuracy effectively evaluates global performance on balanced datasets, it may overestimate model utility when malicious samples are underrepresented.

Area Under the Receiver Operating Characteristic curve (AUROC) quantifies the model’s ability to distinguish positive and negative classes by calculating the integral area under the Receiver Operating Characteristic (ROC) curve. The ROC curve plots True Positive Rate (TPR) against False Positive Rate (FPR):

$$TPR = \frac{TP}{TP+FN}$$

$$FPR = \frac{FP}{FP+TN}$$

F1 Score harmonizes Precision and Recall through their harmonic mean:

$$Precision = \frac{TP}{TP+FP}$$

$$Recall = \frac{TP}{TP+FN}$$

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

Given the asymmetric costs of false positives (over-censorship) and false negatives (hate propagation) in real-world moderation, we prioritize Malicious-Class F1 $F1_{\text{Malicious}}$ and additionally report Macro-F1 for class-balanced assessment.

4.3 Ablation Study

In this section, we meticulously conducted an ablation study to highlight the significant advantages of the LaMHA model. We selected PyTorch as the implementation framework and performed model training on hardware equipped with an RTX 4090 graphics card, ensuring the efficiency of the experiment. To ensure consistency and comparability of experimental results, all models compared were trained and tested on the HarMeme dataset and the HMC dataset, with a fine-tuned learning rate of 0.00001 and a batch size of 256. In all experiments, we utilized ViT-L/14 as the backbone network for CLIP [31]. The entire system was trained using the standard binary cross-entropy loss function.

As demonstrated in Table 2, the LaMHA model (LoRA + aMLP) achieves significant performance gains in multimodal malicious meme classification. On the HMC dataset, it improves F1-score by 11.53% (58.26→69.79) and AUROC by 1.85 points (82.16→84.01) over the Base model. For the more semantically complex HarMeme dataset, accuracy (Acc) and AUROC increase by 4.49% (79.23→83.72) and 7.85% (84.31→92.16), respectively. Key findings include: 1) Using LoRA alone elevates the F1 by 5.27% (69.85→75.12) on HarMeme dataset, proving low-rank adaptation effectively resolves implicit semantic alignment; 2) Integrating aMLP further boosts F1 by 4.7% (75.12→79.82), confirming dynamic gated fusion optimizes cross-modal decision boundaries. The synergistic gains (HMC F1 +3.46%, HarMeme Acc +5.07%) surpass the sum of individual contributions, underscoring the superiority of the “alignment-discrimination” cascaded architecture.

Table 2 further reveals module-specific adaptability: LoRA excels on culture-dependent HarMeme (AUROC +4.33% vs. HMC’s +0.35%) as low-rank constraints capture fine-grained cultural metaphors, while aMLP rectifies LoRA’s overfitting tendency on HMC dataset (Acc 76.14→77.74) via attention gating against adversarial noise. Efficiency-wise, LoRA tunes only 0.3% of CLIP text encoder parameters (34.9K/11.8M) to deliver 90% of full-parameter tuning performance; aMLP increases inference throughput 3.2× (5.2→16.7 samples/ms) with 1.2M parameters. This “efficient projection to dynamic rectification” paradigms pioneers’ real-time multimodal moderation.

4.4 Comparison with State-of-the-Art method

As shown in Table 3, LaMHA demonstrates balanced and state-of-the-art performance in multimodal meme classification. On the HMC dataset, LaMHA achieves 77.74% accuracy (Acc) and 69.79% F1-score, surpassing all baselines (e.g., ISSUES: 77.70% Acc/68.91% F1) with marginal yet consistent improvements (+0.04% Acc and +0.88% F1). For the more challenging HarMeme dataset, LaMHA sets a new AUROC record of 92.16% (+0.18% over ISSUES), validating its strong capability in decoding cultural metaphors and implicit attacks. Although it exhibits a slightly lower F1-score (79.82%)

Table 2: Results of ablation experiments performed on the HMC dataset & HarMeme dataset.

Dataset	Method	LoRA	aMLP	Acc	F1	AUROC
HMC	Base			75.15	58.26	82.16
		✓		76.14	66.33	82.51
HarMeme	LaMHA	✓	✓	77.74	69.79	84.01
	Base			79.23	69.85	84.31
		✓		78.65	75.12	88.64
	LaMHA	✓	✓	83.72	79.82	92.16

Table 3: Comparison with existing methods on the HMC dataset & HarMeme dataset.

Method	HMC dataset			HarMeme dataset		
	Acc.	F1	AUROC	Acc.	F1	AUROC
CLIP Text-Only	63.50	52.36	63.43	73.79	71.60	79.31
CLIP Image-Only	74.65	57.63	81.35	79.38	78.66	88.54
ViT-L/14	70.03	58.94	72.78	77.27	75.55	85.53
CLIP [31]	76.80	64.87	82.92	81.36	80.30	87.27
ISSUES [33]	77.70	68.91	85.51	81.31	80.45	91.98
Hate-CLIPper [30]	77.15	67.59	84.36	83.68	83.31	90.83
LaMHA	77.74	69.79	84.01	83.72	79.82	92.16

compared to Hate-CLIPper (83.31%) on HarMeme dataset, Table 3 reveals its superior cross-modal robustness and adversarial sample adaptability (e.g., +2.2% F1 over CLIP-Image only on HMC dataset), confirming the synergistic benefits of dynamic gating (aMLP) and low-rank adaptation (LoRA).

Table 3 further highlights critical insights: 1) On HMC dataset, LaMHA improves Acc by 3.09% over vision-only CLIP-Image (74.65%→77.74%), proving the necessity of cross-modal fusion; 2) On HarMeme dataset, LaMHA significantly outperforms unimodal methods in AUROC (92.16% vs. CLIP-Text’s 79.31% and CLIP-Image’s 88.54%), demonstrating that dynamic cross-modal interaction effectively captures implicit semantic correlations. Current limitations include sensitivity to vision-dominant samples (e.g., 3.49% F1 gap on HarMeme). Future work will optimize aMLP’s spatial attention weighting and enhance generalization capability.

5 CONCLUSION

This paper proposes a novel multimodal malicious meme classification architecture based on Low-Rank Adaptation (LoRA) and Attention-guided Multilayer Perceptron (aMLP), which achieves efficient and accurate cross-modal semantic parsing through a cascaded “lightweight alignment → dynamic discrimination” design. Key findings include:

- The LoRA module significantly enhances cross-modal feature alignment via low-rank projection, achieving a state-of-the-art AUROC of 92.16% on the culture-dependent HarMeme dataset, surpassing the previous best method (ISSUES) by 0.18%.
- The dynamic gating mechanism in aMLP effectively suppresses noise interference, improving the F1score to 69.79% on adversarial sample-rich HMC dataset, a 17.43% gain over unimodal baselines (CLIP-Text only).

- The synergistic effect of both modules generates nonlinear performance gains, demonstrating consistent robustness in cross-dataset evaluations (superior Accuracy on both HMC and HarMeme datasets). While this architecture pioneers a technical pathway for real-time multimodal content moderation, its sensitivity to vision-dominant samples requires further optimization. Future work will explore spatially aware attention gating to enhance cultural generalization.

6 DISCLAIMER

This paper contains discriminatory content that may be disturbing to some readers.

References

- [1] Christian Bauckhage. 2021. Insights into Internet Memes. Proceedings of the International AAAI Conference on Web and Social Media 5, 1 (August 2021), 42–49. <https://doi.org/10.1609/icwsm.v5i1.14097>
- [2] Richard Dawkins. 2006. The Selfish Gene: 30th Anniversary edition. OUP Oxford.
- [3] Douwe Kiela, Hamed Firooz, Aravind Mohan, Vedanuj Goswami, Amanpreet Singh, Pratik Ringshia, and Davide Testuggine. 2020. The hateful memes challenge: detecting hate speech in multimodal memes. In Proceedings of the 34th International Conference on Neural Information Processing Systems (NIPS ’20). Curran Associates Inc., Red Hook, NY, USA, Article 220, 2611–2624.
- [4] Siddhant Bikram Shah, Shuvam Shiwakoti, Maheep Chaudhary, and Haohan Wang. 2024. MemeCLIP: Leveraging CLIP Representations for Multimodal Meme Classification. Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (2024), 17320–17332. <https://doi.org/10.18653/v1/2024.emnlp-main.959>
- [5] Amanda Williams, Clio Oliver, Katherine Aumer, and Chanel Meyers. 2016. Racial microaggressions and perceptions of Internet memes. Comput. Hum. Behav. 63, C (October 2016), 424–432. <https://doi.org/10.1016/j.chb.2016.05.067>
- [6] Jessica Drakett, Bridgette Rickett, Katy Day, Kate Milnes, 2018. Old Jokes, New Media – Online Sexism and Constructions of Gender in Internet Memes. Feminism & Psychology. 28, 1 (Feb. 2018), 109–127.
- [7] Imperato, Chiara, Maria Pagano, and Tiziana Mancini. 2023. “All Is Fair in . . . Meme!” How Heterosexual Users Perceive and React to Memes, News, and Posts Discriminating against Sexual Minorities” Social Sciences 12, no. 2: 74. <https://doi.org/10.3390/socsci12020074>
- [8] Surendrabikram Thapa, Kritesh Rauniyar, Shuvam Shiwakoti, Sweta Poudel, Usman Naseem, and Mehwish Nasim. 2023. NEHATE: Large-Scale Annotated

- Data Shedding Light on Hate Speech in Nepali Local Election Discourse. *Frontiers in Artificial Intelligence and Applications* (September 2023). <https://doi.org/10.3233/faia230535>
- [9] Stijn Peeters, Marc Tuters, Tom Willaert, and Daniël de Zeeuw. 2021. On the Vernacular Language Games of an Antagonistic Online Subculture. *Front. Big Data* 4, (August 2021). <https://doi.org/10.3389/fdata.2021.718368>
- [10] Kyle Langvardt. 2017. Regulating online content moderation. *Geo. LJ*, 106:1353.
- [11] Chhavi Sharma, Deepesh Bhageria, William Scott, Srinivas PYKL, Amitava Das, Tanmoy Chakraborty, Viswanath Pulabaigari, and Björn Gambäck. 2020. SemEval-2020 Task 8: Memotion Analysis- the Visuo-Lingual Metaphor! *Proceedings of the Fourteenth Workshop on Semantic Evaluation* (2020). <https://doi.org/10.18653/v1/2020.semeval-1.99>
- [12] Shraman Pramanick, Dimitar Dimitrov, Rituparna Mukherjee, Shivam Sharma, Md. Shad Akhtar, Preslav Nakov, and Tanmoy Chakraborty. 2021. Detecting Harmful Memes and Their Targets. *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021* (2021). <https://doi.org/10.18653/v1/2021.findings-acl.246>
- [13] Shraman Pramanick, Shivam Sharma, Dimitar Dimitrov, Md. Shad Akhtar, Preslav Nakov, and Tanmoy Chakraborty. 2021. MOMENTA: A Multimodal Framework for Detecting Harmful Memes and Their Targets. *Findings of the Association for Computational Linguistics: EMNLP 2021* (2021). <https://doi.org/10.18653/v1/2021.findings-emnlp.379>
- [14] Suryawanshi, Shardul and Chakravarthy, Bharathi Raja and Arcan, Mihael and Buitelaar, Paul. 2020. Multimodal meme dataset (MultiOFF) for identifying of-fensive content in image and text. In *Proceedings of the second workshop on trolling, aggression and cyberbullying*. 32–41.
- [15] Vishal Keswani, Sakshi Singh, Suryansh Agarwal, and Ashutosh Modi. 2020. IITK at SemEval-2020 Task 8: Unimodal and Bimodal Sentiment Analysis of Internet Memes. *Proceedings of the Fourteenth Workshop on Semantic Evaluation* (2020), 1135–1140.
- [16] Chen Ling, Ihab AbuHilal, Jeremy Blackburn, Emiliano De Cristofaro, Savvas Zannettou, and Gianluca Stringhini. 2021. Dissecting the meme magic: Understanding indicators of virality in image memes. *Proceedings of the ACM on human-computer interaction* 5, CSCW1 (2021), 1–24.
- [17] Tanaka, Kohtaro and Yamane, Hiroaki and Mori, Yusuke and Mukuta, Yusuke and Harada, Tatsuya. 2022. Learning to evaluate humor in memes based on the incongruity theory. In *Proceedings of the Second Workshop on When Creative AI Meets Conversational AI*, pages 81–93.
- [18] Jingnong Qu, Liunan Harold Li, Jieyu Zhao, Sunipa Dev, and Kai-Wei Chang. 2022. DisinfoMeme: A Multimodal Dataset for Detecting Meme Intentionally Spreading Out Disinformation. (2022). <https://doi.org/10.48550/ARXIV.2205.12617>
- [19] Aashish Bhandari, Siddhant B. Shah, Surendrabikram Thapa, Usman Naseem, and Mehwish Nasim. 2023. CrisisHateMM: Multimodal Analysis of Directed and Undirected Hate Speech in Text-Embedded Images from Russia-Ukraine Conflict. 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW) (June 2023), 1994–2003. <https://doi.org/10.1109/cvprw59228.2023.00193>
- [20] Victoria Sherratt. 2022. Towards Contextually Sensitive Analysis of Memes: Meme Genealogy and Knowledge Base. *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence* (July 2022), 5871–5872. <https://doi.org/10.24963/ijcai.2022/838>
- [21] Sethi, Anushka, Utkarsh Kuchhal, Anjum, and Rahul Katarya. 2021. "Study of Various Techniques for the Classification of Hateful Memes," *International Conference on Recent Trends on Electronics, Information, Communication & Technology (RTEICT)*, Bangalore, India, 2021, pp. 675–680.
- [22] Bhat, Aruna, Vaibhav Vashisht, Vaibhav Raj Sahni, and Sumit Meena. 2023. "Hate Speech Detection using Multimodal Meme Analysis," 2nd International Conference on Applied Artificial Intelligence and Computing (ICAIC), Salem, India, 2023, pp. 1137–1142.
- [23] Huaicheng Fang, Fuqing Zhu, Jizhong Han, and Songlin Hu. 2022. Multimodal Hateful Memes Detection via Image Caption Supervision. 2022 IEEE Smartworld, Ubiquitous Intelligence & Computing, Scalable Computing & Communications, Digital Twin, Privacy Computing, Metaverse, Autonomous & Trusted Vehicles (December 2022), 1530–1537. <https://doi.org/10.1109/smartworld-uic-atc-scalcom-digitaltwin-pricomp-metaverse56740.2022.00221>
- [24] Nikhil Rasiwasia, Jose Costa Pereira, Emanuele Coviello, Gabriel Doyle, Gert R.G. Lanckriet, Roger Levy, and Nuno Vasconcelos. 2010. A new approach to cross-modal multimedia retrieval. In *Proceedings of the 18th ACM international conference on Multimedia (MM '10)*. Association for Computing Machinery, New York, NY, USA, 251–260. <https://doi.org/10.1145/1873951.1873987>
- [25] Stefanie Schmidt and Wolfgang G. Stock. 2009. Collective indexing of emotions in images. A study in emotional information retrieval. *Journal of the American Society for Information Science and Technology* 60, 5 (February 2009), 863–876. <https://doi.org/10.1002/asi.21043>
- [26] Donglin Cao, Rongrong Ji, Dazhen Lin, and Shaozi Li. 2014. A cross-media public sentiment analysis system for microblog. *Multimedia Systems* 22, 4 (August 2014), 479–486. <https://doi.org/10.1007/s00530-014-0407-8>
- [27] Yuhai Yu, Hongfei Lin, Jiana Meng, and Zhehuan Zhao. 2016. Visual and Textual Sentiment Analysis of a Microblog Using Deep Convolutional Neural Networks. *Algorithms* 9, 2 (June 2016), 41. <https://doi.org/10.3390/a9020041>
- [28] Xingyue Chen, Yunhong Wang, Qingjie Liu. 2017. Visual and textual sentiment analysis using deep fusion convolutional neural networks. In *Proceedings of the 2017 IEEE International Conference on Image Processing (ICIP)* 1557–1561.
- [29] Quanzeng You, Jiebo Luo, Hailin Jin, and Jianchao Yang. 2015. Joint Visual-Textual Sentiment Analysis with Deep Neural Networks. In *Proceedings of the 23rd ACM international conference on Multimedia (MM '15)*. Association for Computing Machinery, New York, NY, USA, 1071–1074. <https://doi.org/10.1145/2733373.2806284>
- [30] Gokul Karthik Kumar and Karthik Nandakumar. 2022. Hate-clipper: Multimodal hateful meme classification based on cross-modal interaction of clip features. *arXiv preprint arXiv:2210.05916*.
- [31] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger and Ilya Sutskever. 2021. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*. PMLR, 8748–8763.
- [32] Mei Jingbiao, Jinghong Chen, Weizhe Lin, Bill Byrne, and Marcus Tomalin. 2023. Improving Hateful Meme Detection through Retrieval-Guided Contrastive Learning. *arxiv preprint arxiv:2311.08110*.
- [33] Burbi, Giovanni, Alberto Baldrati, Lorenzo Agnolucci, Marco Bertini, and Alberto Del Bimbo. 2023. Mapping memes to words for multimodal hateful meme classification. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 2832–2836.
- [34] Hu, Edward J., Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. Lora: Low-rank adaptation of large language models. *ICLR* 1, no. 2: 3.
- [35] Liu, Hanxiao, Zihang Dai, David So, and Quoc V. Le. 2021. Pay attention to MLPs. *Advances in neural information processing systems* 34: 9204–9215.
- [36] Pramanick, Shraman, Dimitar Dimitrov, Rituparna Mukherjee, Shivam Sharma, Md Shad Akhtar, Preslav Nakov, and Tanmoy Chakraborty. 2021. "Detecting harmful memes and their targets." *arxiv preprint arxiv:2110.00413*.